



Varendra University
Department of Computer Science and Engineering

Undergraduate Thesis/Project Proposal

Title of the Thesis/Project
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Submitted by:

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Program: B.Sc. in Computer Science and Engineering

Semester: 7th

Section: A

Course Code: CSE 418

Course Title: Project or thesis with Seminar part I

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Submission Date: 24/08/2025

1. Introduction

Courier services are everywhere in Bangladesh, but if we look closely, most of them are built for long-distance or nationwide deliveries. For example, Sundarban or Pathao Courier work well if someone wants to send a parcel from Rajshahi to Dhaka. But what if a student in Rajshahi urgently needs to send documents across the city within an hour? Or a shopkeeper wants to send a small package quickly to a customer nearby? For these short-distance, instant deliveries, there is no dedicated, lightweight system in Rajshahi right now.

Food delivery apps like Foodpanda or Foodi do show real-time tracking, but they only serve restaurants, not general parcels. As a result, people are forced to depend on either manual arrangements or wait for traditional couriers that are slow and inconvenient. This is the gap we want to solve.

Our project aims to develop a simple mobile app, using Flutter (for Android & iOS) and Firebase as backend, that connects customers, delivery men, and admin in a single system. Customers will be able to book a parcel within Rajshahi city and track its status through a timeline system. Delivery men can accept tasks and update statuses, while the admin will monitor everything.

We also plan to include SSLCommerz payment integration, so customers can pay online securely instead of depending only on cash.

This project is not about competing with big courier companies, but about filling a local need. It will give us hands-on experience in real-world mobile app development while also creating value for the Rajshahi community by making deliveries faster, more reliable, and more transparent.

In addition, the system will integrate direct communication features such as calling and messaging so that customers can easily contact delivery men. Using Google Maps integration, customers will also be able to view the nearest available delivery man before placing an order. This ensures faster response time and improves reliability.

2. Background Study

- **Pathao Courier (Bangladesh):** Provides nationwide service but lacks localized instant delivery features for smaller cities.
- **Foodi / Foodpanda:** Strong in real-time food delivery but not designed for small parcel delivery.
- **Delhivery (India):** A large-scale logistics system focused on businesses rather than fast, city-level courier services.

3. Problem Statement

Courier services in Rajshahi lack a proper technology-based platform for quick parcel delivery. Customers face:

- No proper mobile system to book and manage deliveries.
- No transparency in parcel status.
- Delays in delivery due to inefficient processes.

Thus, there is a need for a city-level courier system that is fast, reliable, and transparent.

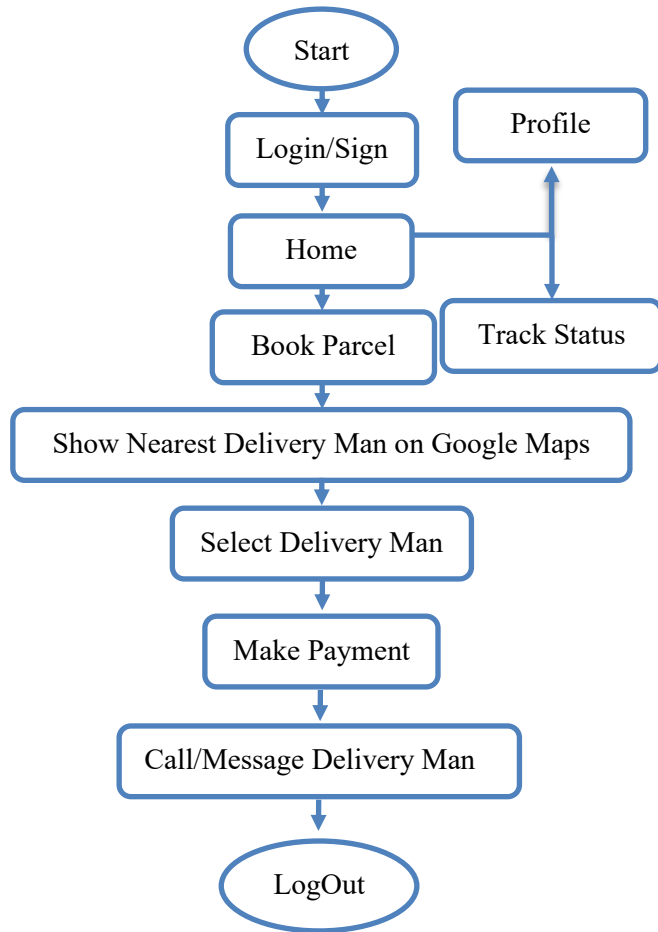
4. Objectives

- Develop a cross-platform courier service mobile application using Flutter and Firebase.
- Implement a timeline-based delivery status update system for customers.
- To add Google Maps integration for finding the nearest delivery man and enabling live location sharing.
- To include calling and messaging features for direct communication between customers and delivery men.
- Integrate three user modes: Customer, Delivery Man, and Admin.
- Needs to be a skilled delivery man who can complete the job within the specified time.

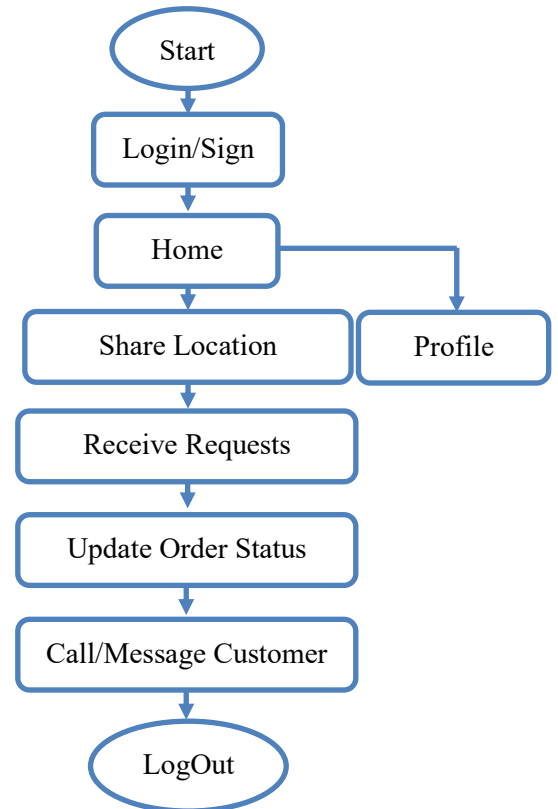
5. System Design

- Architecture: Three-tier system (Frontend – Backend – Database).
- Frontend: Flutter-based mobile app (Android & iOS).
- Backend: Firebase Authentication, Firestore Database, Cloud Functions (Node.js/JavaScript).
- Modules:
 - Customer Module: Register & login, book delivery, View nearest delivery man on Google Maps, Call or message delivery man directly, online payment through SSLCommerz, view status timeline, Profile management.
 - Delivery Man Module: Register & login, Receive delivery requests, Share live location on Google Maps, update parcel status, Call or message customer.
 - Admin Module: Monitor requests, assign deliveries, manage users, Approve driver registrations.

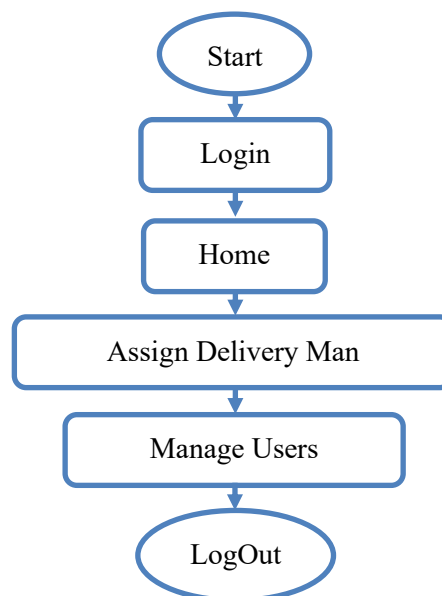
User Flow Chart



Delivery Man Flow Chart



Delivery Man Flow Chart



6. Expected Outcomes

- A working prototype mobile app for Rajshahi courier service.
- Timeline-based parcel tracking system.
- Google Maps integration for locating the nearest delivery man and sharing live location.
- Calling and messaging features for smooth communication.
- Admin panel for monitoring and management.
- Scalable system architecture for future city-wide expansion.
- Integrated secure payment system via SSLCommerz.

7. Tools and Technologies to Be Used

- Programming Language: Dart (Flutter)
- Framework: Flutter SDK
- Database: Firebase Firestore
- Services: Firebase Authentication, Cloud Functions, Notifications
- Tools: Android Studio, Xcode, Visual Studio Code, GitHub
- Payment Gateway: SSLCommerz
- APIs: Google Maps API
- Native Features: Phone dialer & SMS integration (using Flutter `url_launcher`).

8. References

- Flutter Documentation [<https://docs.flutter.dev/>]
- Firebase Documentation [<https://firebase.google.com/>]
- Sslcommerz Documentation
[<https://developer.sslcommerz.com/doc/v4/>]
- Pathao Courier Service. [<https://pathao.com>]
- Foodpanda Bangladesh. [<https://foodpanda.com.bd>]
- Delhivery Logistics, India. [<https://www.delhivery.com>]

9. Work Division and Team Management

Team Member	Role	Tasks
Md. Abdul Momin	Designer & Frontend Developer	UI/UX design, prototyping, Flutter coding
Md. Abdullah	Frontend & Backend Developer	Flutter coding, Firebase integration, testing
Md. Zahid Hasan	Planner & Documentation Lead &Frontend	Planning, task scheduling, Flutter coding

10. Timeline

Month	Task / Deliverables
Month 1	Onboarding screens, user login & signup pages, Firebase authentication & database connection
Month 2	Customer home page, order placement page, SSLCommerz payment integration
Month 3	Order timeline (Ordered → Picked → On the Way → Delivered), customer profile page
Month 4	Driver registration & login pages, database integration for driver authentication
Month 5	Driver homepage, order receiving system, delivery status update (Picked/Delivered)
Month 6	Google Maps integration (nearest driver search, live location sharing)
Month 7	Calling & Messaging feature between customer and driver
Month 8	Admin login, admin dashboard, driver registration approval system, driver profile management
Month 9	Additional features, UI/UX optimization, system performance improvements
Month 10	Real-time testing on devices, collect user reviews, final bug fixing & documentation

11. Cost-Benefit Analysis

- Firebase (Free tier sufficient).
- Flutter & Android Studio (Free).
- Hosting (not required in first phase).
- Conclusion: Low-cost, student-friendly, cost-effective.
- SSLCommerz provides affordable student-level integration with per-transaction charges. As payment will be online, it reduces the risk of cash handling and improves transparency.
- Google Maps API: Free for limited usage; costs may apply if scaling up, but manageable within student-level limits.

12. Constraints and Limitations

- Limited to Rajshahi city only in initial phase.
- Requires internet for real-time updates.
- Free tiers of Firebase and Google Maps may have usage limitations.

13. Societal Impact

- Provides faster, more reliable courier service in Rajshahi.
- Improves trust and transparency in parcel handling.
- Creates employment opportunities for delivery personnel.
- Promotes digital payment culture in Rajshahi, reduces cash dependency, and aligns with Bangladesh's vision of Digital Financial Inclusion.

14. Complex Engineering Problem Mapping

Fill this table by reflecting how your proposal maps with the complex engineering problem domains.

Table 1: Mapping with complex problem solving.

WP1 Dept of Knowledge	WP2 Range of Conflicting Requirements	WP3 Depth of Analysis	WP4 Familiarity of Issues	WP5 Extent of Applicable Codes	WP6 Extent of Stakeholder Involvement	WP7 Interdependence
✓	✓					

15. Complex Engineering Activity Mapping

Justify which complex engineering activity level your proposal aligns with.

Table 2: Mapping with complex engineering activities.

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓			

16. Green Computing Considerations

- Cloud-based Firebase reduces physical infrastructure needs.
- Eliminates paper-based courier logs.
- Optimized system reduces wasted trips and saves fuel.

17. Lifelong Learning and Personal Development

- Improves skills in Flutter, Firebase, Google Maps, and mobile app development.
- Builds teamwork, project management, and communication abilities.
- Strengthens problem-solving with real-world delivery challenges.
- Provides hands-on experience with secure payments and cloud services.
- Prepares for the software industry and future technologies.